

Module 6: Multivariable & Constrained Optimization

Ph.D. Math Camp

1 Partial Differentiation

Often, the functions we are interested in are function of more than a single variable. For example, in microeconomics Q =output is often a function of capital (K) and labor (L).

Let's consider a function

$$y = f(x_1, x_2, \dots, x_n)$$

where $x_1, x_2, \dots, x_n$ are independent of each other. If x_1 undergoes a change Δx_1 , while $x_2, \dots, x_n$ remain fixed, there will be a corresponding change in y , Δy .

We can express this using our old secant formula

$$\frac{\Delta y}{\Delta x_1} = \frac{f(x_1 + \Delta x_1, x_2, \dots, x_n) - f(x_1, x_2, \dots, x_n)}{\Delta x_1}$$

If we take the limit as Δx_1 approaches zero $\implies$ a derivative. Which we call the **partial derivative of y with respect to x_1** .

This indicates all other variables will be held constant. We use the following notation $\frac{\partial y}{\partial x_1}$. Read "partial f, partial x one."

Instead of f' we use f_1 or f_x , where the subscript indicates the variable we are taking the partial w.r.t.

Example: Find the partial derivatives of $f(x_1, x_2) = 3x_1^2 + x_1x_2 + 4x_2^2$

$$f_1 = 6x_1 + x_2$$

$$f_2 = x_1 + 8x_2$$

Note that the partials are functions of x_1 and x_2 themselves which means they can be evaluated. For instance at a point like $(x_1, x_2) = (1, 3)$.

Examples: Find partials

1. $z = 5x^3 - 3x^2y^2 + 7y^5$

2. $f = xe^{-2xy}$

3. $z = 4x^3 + 2x^2y - 7y^5$

All normal differentiation rules apply.

1.1 Geometric Interpretation of Partials

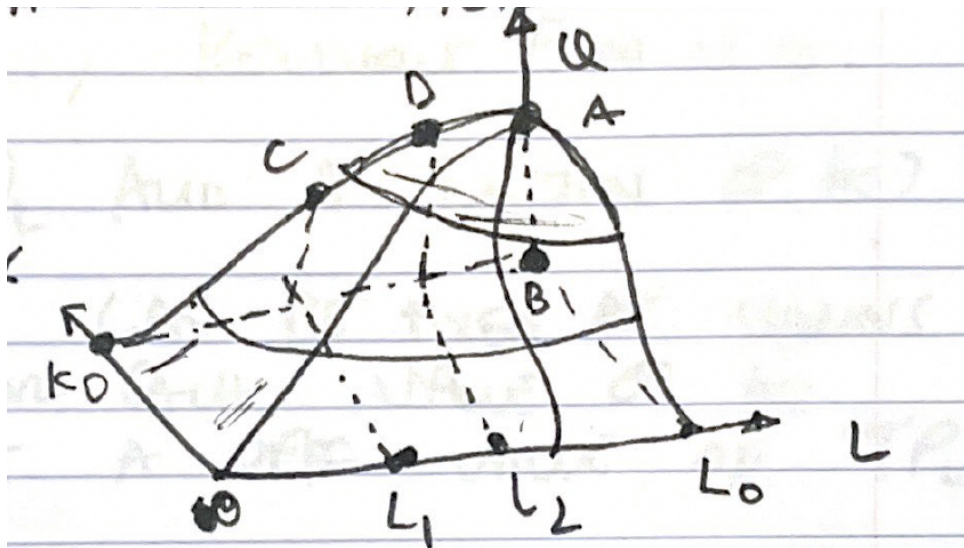
Given that the partial is a type of derivative it has a slope interpretation.

Consider $Q = Q(K, L) \implies$ we can define 2 partials $\frac{\partial Q}{\partial K}$ or Q_K and $\frac{\partial Q}{\partial L}$ or Q_L .

Q_K : The rate of change of output with respect to a small change in capital, while labor input is held constant $\implies MP_K$ or **marginal product of K**. Similarly, Q_L , MP_L is the **marginal product of labor**.

We can depict $Q = Q(K, L)$ with a production surface.

Figure 1: K-L Plane



Here, Q is plotted vertically so that for any point (K, L) in the (base) K, L plane the height of the surface will indicate output. The domain should be the non-negative quadrant.

Suppose we hold K fixed at k_0 and consider variations in the input of L . **By setting $K = k_0$, all points in the domain become irrelevant except those on the k_0B line.**

$\Rightarrow$ **Only the curve k_0CDA matters.** This curve represents the total product of labor (TP_L) for a fixed amount of $K = k_0$.

$\Rightarrow$ the slope of curve k_0CDA represents the geometric counterpart of Q_L . **k_0CDA is a cross section of the production surface.**

$\Rightarrow$ slope is MP_L . Note that Q_L is a function of L . When $L = L_1$, the value of Q_L is the slope at point c .

When $L = L_2$, the relevant point is D .

Why is Q_L also a function of K ? Well, K can be fixed at various levels. For each value of k , we have a different value of TP_L .

We can have a **similar discussion** about Q_K . Suppose we held labor constant at

$L_0 \implies L_0B$ is the relevant line segment.

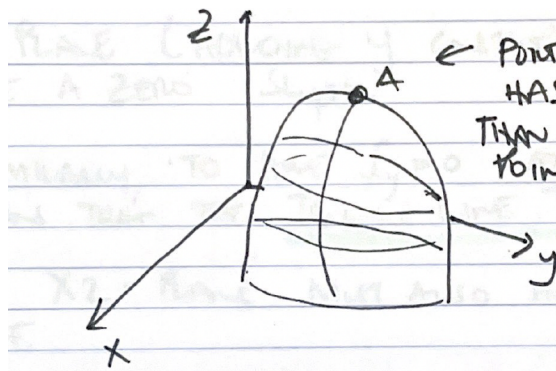
$\implies L_0A$ is the relevant subset of the production surface and Q_K is the slope of the curve L_0A and is a function of L and K .

2 Multivariable Optimization

For a function of one choice variable, extreme values are peaks and valleys.

When we have two choice variables, the graph $z = f(x, y)$ becomes a surface in 3-D space and our peaks and valleys take on a 3-D form $\implies$ a dome or a bowl.

Figure 2: Max in 3-D space



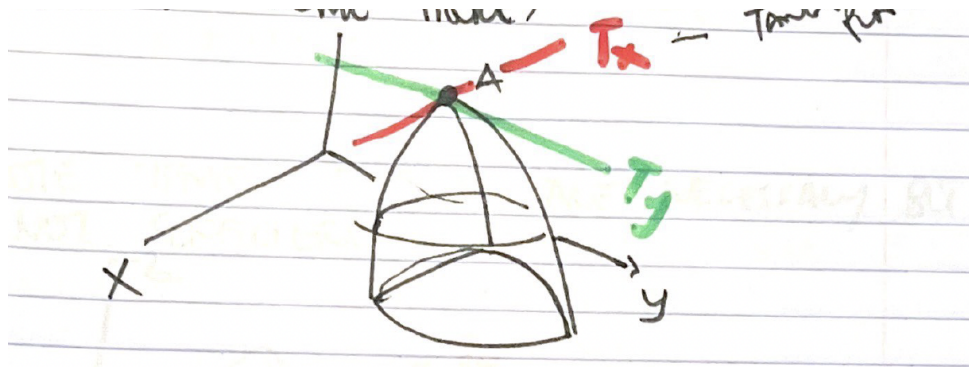
Point A has the largest z value in its neighborhood.

2.1 First-Order Condition

A point in the domain of $z = f(x, y)$ are critical points when

$$f_x = f_y = 0 \quad \text{or} \quad \frac{\partial z}{\partial x} = \frac{\partial z}{\partial y} = 0$$

Figure 3: Tangent Plane



To have $f_x = 0$ at A means tangent line T_x (drawn through A and parallel to xz plane (holding y constant), must have zero slope.

Similarly, to have $f_y = 0$ at point A means that the tangent line T_y , parallel to yz -plane must also have zero slope.

Similar, in case of a minimum.

These F.O.C. or necessary conditions: $f_x = 0$, $f_y = 0$

Example: Find points that satisfy F.O.C. for $z = f(x, y) = 3x^2 - xy + 2y^2 - 4x - 7y + 12$

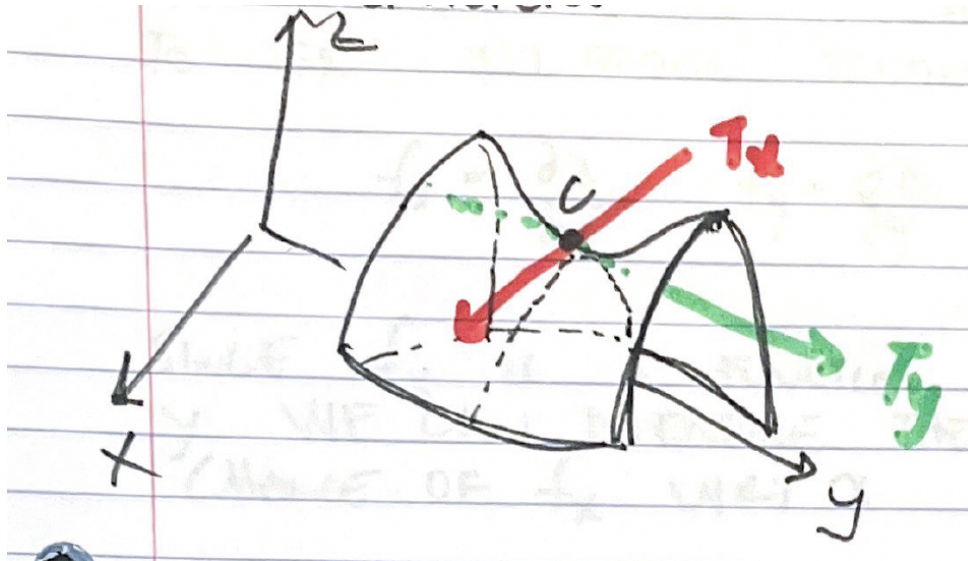
$$f_x = 6x - y - 4 = 0$$

$$f_y = -x + 4y - 7 = 0$$

$$\implies y = 2, x = 1$$

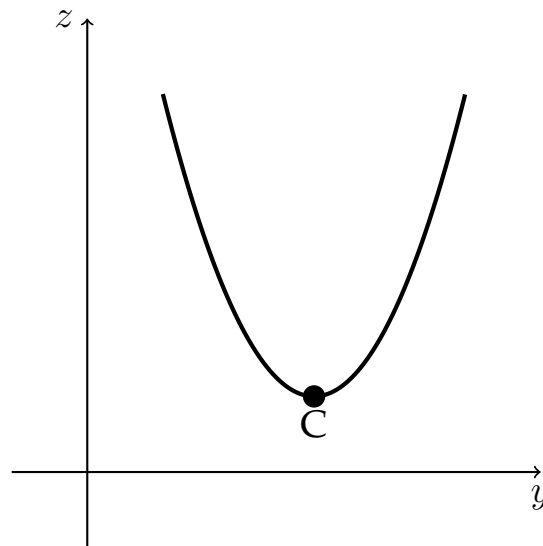
Note that F.O.C. are necessary but not sufficient.

Figure 4: Saddle Point

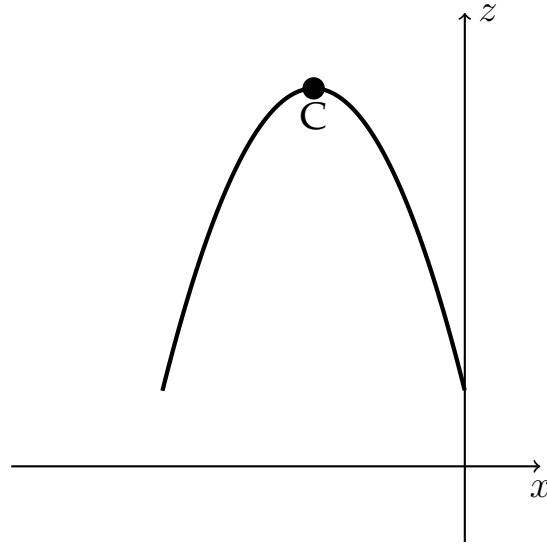


At point C , T_x and T_y have zero slopes, but this point does not qualify as an extremum.

When viewed against the yz -plane you get a **minimum**.



When viewed against the xz -plane you get a **maximum**



To develop a sufficient condition we need second-order partial derivatives.

2.2 Second-Order Conditions

Recall that $z = f(x, y)$ can give rise to two first order partials

$$f_x = \frac{\partial z}{\partial x} \quad \text{and} \quad f_y = \frac{\partial z}{\partial y}$$

Since f_x is a function of x and y we can measure the rate of change of f_x with respect to x and y .

Differentiating with respect to x twice

$$f_{xx} = \frac{\partial}{\partial x} f_x \quad \text{or} \quad \frac{\partial^2 z}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial x} \right)$$

Same for y

$$f_{yy} = \frac{\partial}{\partial y} f_y \quad \text{or} \quad \frac{\partial^2 z}{\partial y^2} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial y} \right)$$

We can also obtain cross (or mixed) partial derivatives

$$f_{xy} = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right)$$

and

$$f_{yx} = \frac{\partial^2 z}{\partial y \partial x} = \frac{\partial}{\partial y} \left(\frac{\partial z}{\partial x} \right)$$

In this case the partial measures the rate of change of one first order partial with respect to the “other” variable. Note: some texts define f_{xy} in the opposite order, as $\frac{\partial}{\partial y}(f_x)$; by Young’s theorem below, the two cross partials are equal for well-behaved functions, so the order does not matter.

Even though f_{xy} and f_{yx} have been separately defined, they **will have identical values** (as long as the cross partials are continuous)– known as *Young’s theorem*.

Also works for functions with 3 or more variables. If $z = g(u, v, w) \implies g_{uv} = g_{vu}$; $g_{vw} = g_{wv}$.

Examples: Find $f_{xx}, f_{xy}, f_{yx}, f_{yy}$

1. $z = f(x, y) = x^3 + 5xy - y^2$

2. $z = f(x, y) = x^2 e^{-y}$

2.3 Interpreting S.O.C.

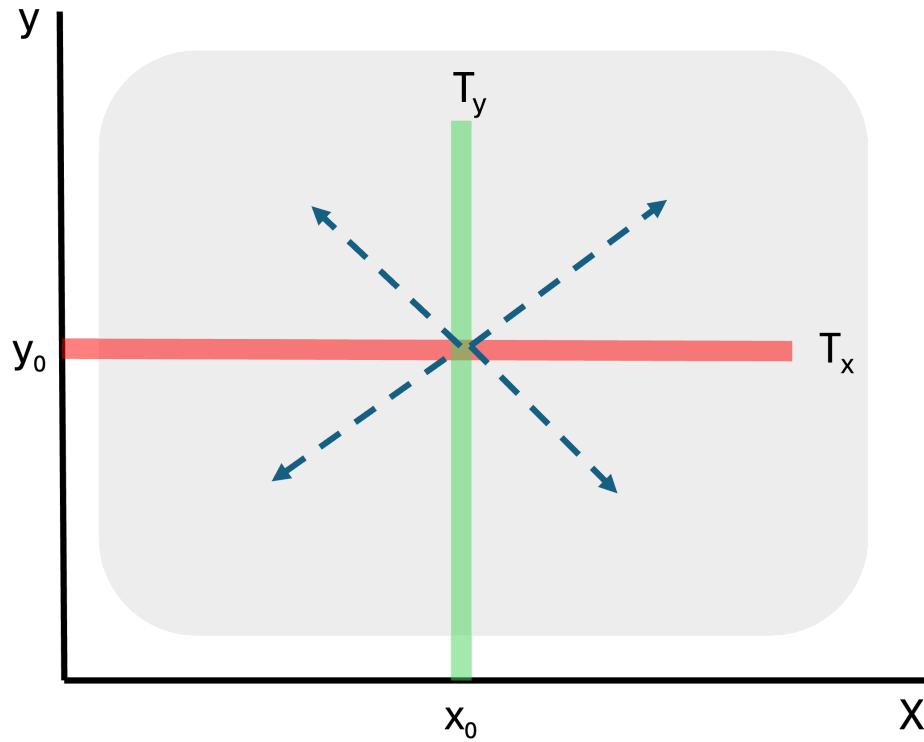
Let $D = f_{xx}f_{yy} - (f_{xy})^2$.

- We will see a max if $f_{xx} < 0, f_{yy} < 0$ and $D > 0 \implies f_{xx}f_{yy} > (f_{xy})^2$
- We get a minimum if $f_{xx} > 0, f_{yy} > 0$ and $D > 0 \implies f_{xx}f_{yy} > (f_{xy})^2$

The S.O.C. depends on f_{xx} and f_{yy} which have to do with the surface configuration around point A.

However, we consider the cross partials as well. Let’s take a bird’s eye view.

Figure 5: Bird's Eye View



The cross partial ensures that the surface that is tangent will yield the same type of configuration (i.e. hill or valley) in any direction (NE, SW, etc.)

Let's summarize with a table

Condition	Max	Min
F.O.C.	$f_x = f_y = 0$	$f_x = f_y = 0$
S.O.C.	$f_{xx}, f_{yy} < 0$ and $f_{xx}f_{yy} > f_{xy}^2$	$f_{xx}, f_{yy} > 0$ and $f_{xx}f_{yy} > f_{xy}^2$

Note that S.O.C are sufficient but not necessary (e.g., $D=0$).

- If $f_{xx}f_{yy} = f_{xy}^2$ we may still have an extremum.
- If $f_{xx}f_{yy} < f_{xy}^2 \implies$ a saddle point e.g., $f_{xx} > 0$ and $f_{yy} < 0$

Example: Find extreme values of $z = 8x^3 + 2xy - 3x^2 + y^2 + 1$

FOC:

$$\begin{aligned}f_x : \quad & 24x^2 + 2y - 6x = 0 \\f_y : \quad & 2x + 2y = 0 \\ \implies x = -y \implies & 24x^2 + 2(-x) - 6x = 0 \\ \implies 24x^2 - 2x - 6x = 0 \implies & 24x^2 - 8x = 0 \\ \implies x(24x - 8) = 0 \\ \implies x_1^* = 0, y_1^* = 0, \quad & x_2^* = \frac{1}{3}, y_2^* = -\frac{1}{3}\end{aligned}$$

SOC at $x = 0, y = 0$

$$\begin{aligned}f_{xx} = 48x - 6 \implies f_{xx} \Big|_{x=0} &= -6 \\f_{yy} &= 2 \\f_{xy} = f_{yx} &= 2\end{aligned}$$

Since $f_{xx}f_{yy} < f_{xy}^2 \implies$ a saddle point $\implies$ surface curls up in one dimension but down in another.

SOC at $x = \frac{1}{3}, y = -\frac{1}{3}$

$$\begin{aligned}f_{xx} = 48x - 6 \implies \Big|_{x=\frac{1}{3}} &= 16 - 6 = 10 \\f_{yy} &= 2 \\f_{xy} = f_{yx} &= 2\end{aligned}$$

Since $f_{xx}f_{yy} > f_{xy}^2$ or $10 \times 2 > (2)^2 \implies$ a **minimum**.

Example: Find extreme values of $z = x + 2e^y - e^x - e^{2y}$

3 Determinant Test for Sign Definiteness

The Second order sufficient condition for extremum stipulates it be **definitely positive** (for a minimum) and **definitely negative** (for a maximum).

Rewriting the second order conditions:

- Positive definite if $f_{xx} > 0$ and $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ is greater than zero.
- Negative definite if $f_{xx} < 0$ and $\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ is greater than zero.

Since f_{xx} and f_{yy} need to have the same sign we get the condition from before.

$\begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$ is known as the discriminant. This is a determinant with second order partial derivatives as its elements.

We call this the **Hessian Determinant** (or Hessian).

$$|H| = \begin{vmatrix} f_{xx} & f_{xy} \\ f_{yx} & f_{yy} \end{vmatrix}$$

3.1 Objective Functions with 2+ Variables

We can no longer graph a function when there are more than two choice variables. However, we can still speak of hyper-surfaces that may be tangent in a way to peaks and domes in this hyperdimensional space.

Consider $f(X_1, X_2, X_3) \implies$ 1st partials f_1, f_2, f_3 and 2nd partials $f_{ij} = \frac{\partial^2 z}{\partial x_i \partial x_j}$ for $i, j = 1, 2, 3$.

By virtue of Young's theorem $f_{ij} = f_{ji}$.

FOC $\implies f_1 = f_2 = f_3 = 0$. Just as in the two-variable case we need that 1st order partial derivative to be zero.

For second order condition, if at a stationary value of z we find that the function (there) is positive definite $\implies$ that value of z is a minimum. Analogously, the negative definiteness is a sufficient condition for a value to be a max.

To determine this we set up our Hessian of second partials

$$|H| = \begin{vmatrix} f_{11} & f_{12} & f_{13} \\ f_{21} & f_{22} & f_{23} \\ f_{31} & f_{32} & f_{33} \end{vmatrix}$$

We can then find the **principal minors**:

$$|H_1| = f_{11}, \quad |H_2| = \begin{vmatrix} f_{11} & f_{12} \\ f_{21} & f_{22} \end{vmatrix}, \quad |H_3| = |H| \quad (1)$$

Note that all leading principal minors must be evaluated at the stationary points where $f_1 = f_2 = f_3 = 0$.

- Z^* is a maximum if $|H_1| < 0; |H_2| > 0; |H_3| < 0 \implies$ negative definiteness
- Z^* is a minimum if $|H_1| > 0; |H_2| > 0; |H_3| > 0 \implies$ positive definiteness

Examples: Find extreme values of following and determine if they are min or max.

1. $z = 2x_1^2 + x_1x_2 + 4x_2^2 + x_1x_3 + x_3^2 + 2$

2. $z = -x_1^3 + 3x_1x_3 + 2x_2 - x_2^2 - 3x_3^2$

Summary

Condition	Max	Min
F.O.C.	$f_1 = \dots = f_n = 0$	$f_1 = \dots = f_n = 0$
S.O.C.	$ H_n > 0$ (< 0) for n even (odd)	$ H_n > 0$ for all n

4 Optimization with a Constraint

We just discussed finding relative extrema of an objective function with 2 or more variables.

However, in that discussion the decision made regarding one variable does not impinge upon the choices of the remaining variables $\rightarrow$ *constrained optimization*.

4.1 Effects of A Constraint

Lets consider a utility function $U = x_1x_2 + 2x_1$. Where consumption (choice) of the number of x_1 and x_2 increase wellbeing, U .

U_1 and U_2 are the partial derivatives and known as marginal utilities.

$$U_1 = \frac{\partial U}{\partial x_1} \quad U_2 = \frac{\partial U}{\partial x_2}$$

$$\implies U_1 = x_2 + 2, \quad U_2 = x_1$$

Since they are both positive (e.g., as x_1 and x_2 increase $\implies U$ increases).

This would imply purchasing an infinite number of both goods.

Realistically, a consumer would face a **budget constraint**.

Suppose the consumer spends \$60 on two goods with prices $p_1 = 4$ and $p_2 = 2$, then their budget line can be expressed as

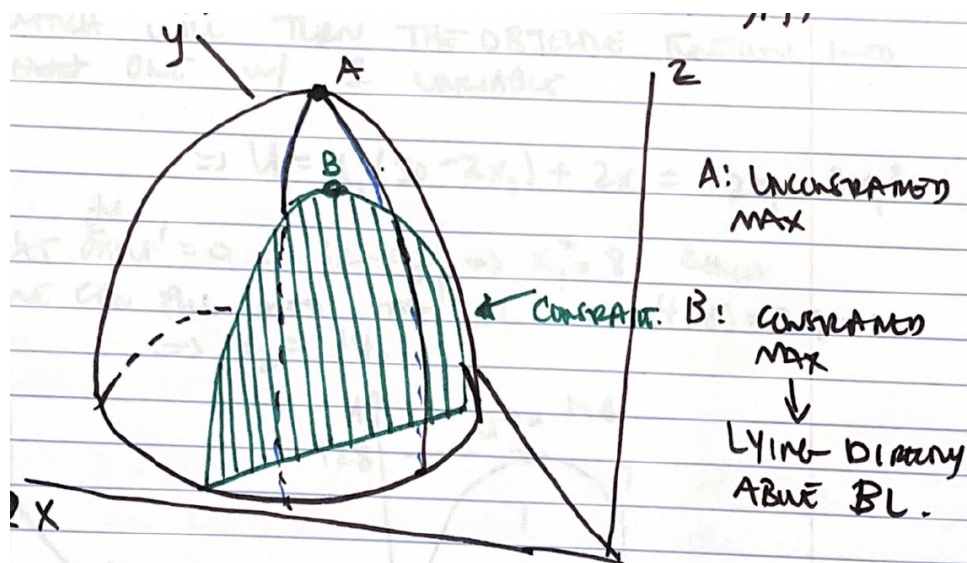
$$4x_1 + 2x_2 = 60$$

Once a BL is introduced, the domain is immediately reduced to the set of points lying on (or below) the budget line.

Only the subset of the utility surface lying directly above the budget constraint line will be relevant.

Generally, for a function $z = f(x, y)$

Figure 6: **Budget Constraint**



Constrained max is usually $\leq$ unconstrained max (although usually less).

Note that if we added another constraint intersecting the first $\implies$ solution is a single point.

$\implies$ usually there would be fewer constraints than choice variables.

4.2 Finding Stationary Values

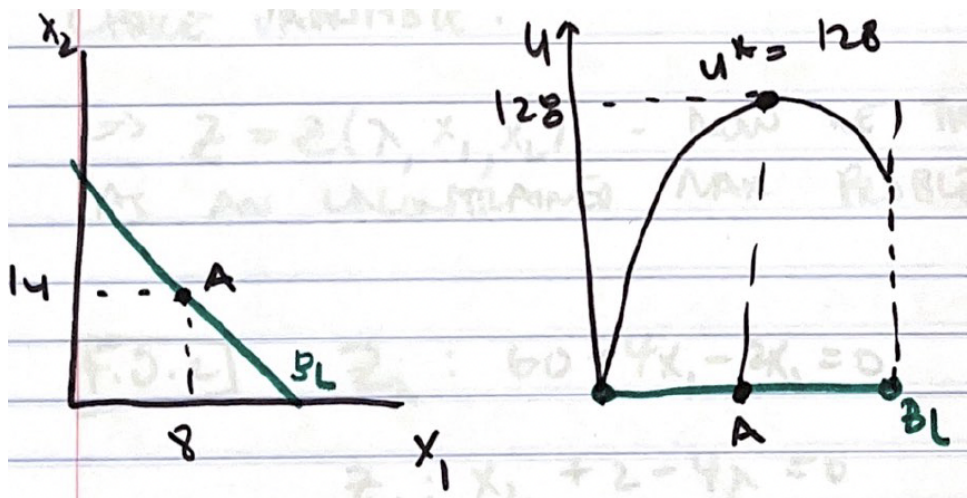
One way to solve the problem

$$\text{Max } U = x_1x_2 + 2x_1 \quad \text{s.t.} \quad 4x_1 + 2x_2 = 60$$

is to substitute $x_2 = 30 - 2x_1$ into the $U(x_1, x_2)$. This turns the objective function into one with 1 variable.

$$\implies U = x_1(30 - 2x_1) + 2x_1 = 32x_1 - 2x_1^2 \quad (2)$$

Set $\frac{dU}{dx} = U' = 0 \implies 32 - 4x_1 \implies x_1^* = 8$. We can then plug this into the BL $\implies 4(8) + 2x_2 = 60 \implies x_2 = 14$.



Since $\frac{d^2U}{dx_1^2} = -4 < 0 \implies$ a max.

If the constraint is more complicated or there are several, this method can become difficult. OR what if we cannot solve for x explicitly?

4.3 Lagrange-Multiplier Method

Using

$$\text{Max } U = x_1x_2 + 2x_1 \quad \text{s.t.} \quad 4x_1 + 2x_2 = 60$$

We can write down the lagrangian function (note how we are adding zero)

$$Z = x_1x_2 + 2x_1 + \lambda(60 - 4x_1 - 2x_2)$$

We can treat λ as an additional choice variable.

We can now treat $Z = Z(\lambda, x_1, x_2)$ as an unconstrained maximization problem.

FOC:

$$Z_\lambda : 60 - 4x_1 - 2x_2 = 0 \quad (1)$$

$$Z_1 : x_2 + 2 - 4\lambda = 0 \quad (2)$$

$$Z_2 : x_1 - 2\lambda = 0 \quad (3)$$

Z_λ guarantees the satisfaction of the constraint even though we treated this as an unconstrained max problem.

The lagrange method allows us to obtain an unconstrained extremum U^* . We can solve for the critical values:

$$\text{From (3)} \implies x_1 = 2\lambda$$

$$\text{From (2)} \implies x_2 = 4\lambda - 2$$

$$\text{From (1)} \implies 60 - 4(2\lambda) - 2(4\lambda - 2) = 0$$

$$\implies 60 - 8\lambda - 8\lambda + 4 = 0 \implies 64 = 16\lambda \implies \lambda = 4$$

$$\implies x_1^* = 8, x_2^* = 14.$$

Generally, for a function $z = f(x, y)$ s.t. $g(x, y) = c$, c is a constant

$$\mathcal{L} = f(x, y) + \lambda[c - g(x, y)]$$

$\implies$

$$\mathcal{L}_\lambda : c - g(x, y) = 0 \quad (1)$$

$$\mathcal{L}_x : f_x - \lambda g_x = 0 \quad (2)$$

$$\mathcal{L}_y : f_y - \lambda g_y = 0 \quad (3)$$

The ingenuity here is that $c - g(x, y)$ is zero so max at $\mathcal{L}$ will be the same as that at Z s.t. $g(x, y) = c$. Since BL is considered in FOC.

Example: $Z = xy$ s.t. $x + y = 6$

4.4 Interpretation of Lagrange Multiplier

It turns out that λ^* is a measure of the sensitivity of Z^* to changes in the constraint.

Given the constraint in Z_λ (FOC) we can think of all the solutions as a function of c . From $Z_\lambda : c - g(x, y) = 0$

$\implies$ the optimal solutions are $\lambda^* = \lambda^*(c), x^* = x^*(c), y^* = y^*(c)$

$$\implies Z^* = f(x^*, y^*) + \lambda^*[c - g(x^*, y^*)]$$

If we differentiate Z^* w.r.t. $c \implies$

$$\frac{dZ^*}{dc} = f_x \frac{dx^*}{dc} + f_y \frac{dy^*}{dc} + \frac{d\lambda^*}{dc} [c - g(x^*, y^*)] + \lambda^* (1 - g_x \frac{dx^*}{dc} - g_y \frac{dy^*}{dc})$$

Now, combine like terms

$$f_x \frac{dx^*}{dc} - \lambda^* g_x \frac{dx^*}{dc} + f_y \frac{dy^*}{dc} - \lambda^* g_y \frac{dy^*}{dc} + \frac{d\lambda^*}{dc} [c - g(x^*, y^*)] + \lambda^*$$

We get

$$(f_x - \lambda^* g_x) \frac{dx^*}{dc} + (f_y - \lambda^* g_y) \frac{dy^*}{dc} + \frac{d\lambda^*}{dc} [c - g(x^*, y^*)] + \lambda^*$$

The first three terms are equal to zero from FOC. $\implies \frac{dZ^*}{dc} = \lambda^*$.

This means that the effect of a change in the constraint via c on optimal value z^* .

4.5 n-Variable and Multiconstraint Cases

Now suppose we have

$$Z = f(x_1, x_2, \dots, x_n) \quad s.t. \quad g(x_1, \dots, x_n) = c$$

$$\mathcal{L} = f(x_1, x_2, \dots, x_n) + \lambda[c - g(x_1, \dots, x_n)]$$

$$\mathcal{L}_\lambda : c - g(x_1, \dots, x_n) = 0$$

$$\mathcal{L}_1 : f_1 - \lambda g_1 = 0$$

$$\mathcal{L}_2 : f_2 - \lambda g_2 = 0$$

.....

$$\mathcal{L}_n : f_n - \lambda g_n = 0$$

If you have two constraints $\mathcal{L} = f(x_1, x_2, \dots, x_n) + \lambda[c - g(x_1, \dots, x_n)] + \mu[d - h(x_1, \dots, x_n)]$ you add $\mathcal{L}_\mu$. Solve for all x_i^* and λ^*, μ^* .

Example: $z = xy \quad s.t. \quad x + 2y = 2$

4.6 Second Order Conditions

Now we must adjust what we did for the unconstrained case. Before we used the Hessian now we need a bordered Hessian which we write as follows:

$$|\bar{H}| = \begin{vmatrix} 0 & g_x & g_y \\ g_x & Z_{xx} & Z_{xy} \\ g_y & Z_{yx} & Z_{yy} \end{vmatrix}$$

If $|\bar{H}| < 0 \implies d^2z$ **positive definite subject to the constraint** $\implies$ **a minimum.**

If $|\bar{H}| > 0 \implies d^2z$ **negative definite subject to the constraint** $\implies$ **a maximum.**

Note that all derivatives in $|\bar{H}|$ have to be evaluated at the critical values.

Example: $z = xy$ s.t. $x + y = 6$

$$\mathcal{L} = xy + \lambda(6 - x - y)$$

$$\mathcal{L}_\lambda : 6 - x - y = 0$$

$$\mathcal{L}_x : y - \lambda = 0$$

$$\mathcal{L}_y : x - \lambda = 0$$

$$\implies x^* = y^* = \lambda^* = 3$$

$$|\bar{H}| = \begin{vmatrix} 0 & g_x & g_y \\ g_x & Z_{xx} & Z_{xy} \\ g_y & Z_{yx} & Z_{yy} \end{vmatrix} = \begin{vmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{vmatrix}$$

$$= -(-1) + (1 - 0) = 2 > 0 \implies \text{a max at } (x, y, z) = (3, 3, 9)$$

4.6.1 n-Variable Case

When the objective function takes the form

$$Z = f(x_1, x_2, \dots, x_n) \quad \text{s.t.} \quad g(x_1, \dots, x_n) = c$$

The condition FOR positive or negative definiteness involves a bordered hessian.

We can express this in terms of **bordered leading principal minors.**

$$|\bar{H}| = \begin{vmatrix} 0 & g_1 & g_2 & \cdots & g_n \\ g_1 & Z_{11} & Z_{12} & \cdots & Z_{1n} \\ g_2 & Z_{21} & Z_{22} & \cdots & Z_{2n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ g_n & Z_{n1} & Z_{n2} & \cdots & Z_{nn} \end{vmatrix}$$

Its bordered leading principal minors can be defined as

$$|\bar{H}_2| = \begin{vmatrix} 0 & g_1 & g_2 \\ g_1 & Z_{11} & Z_{12} \\ g_2 & Z_{21} & Z_{22} \end{vmatrix}, \quad |\bar{H}_3| = \begin{vmatrix} 0 & g_1 & g_2 & g_3 \\ g_1 & Z_{11} & Z_{12} & Z_{13} \\ g_2 & Z_{21} & Z_{22} & Z_{23} \\ g_3 & Z_{31} & Z_{32} & Z_{33} \end{vmatrix}$$

The last being $|\bar{H}_n| = |\bar{H}|$.

We have **positive definiteness (a minimum)** if $|\bar{H}_2|, |\bar{H}_3|, \dots, |\bar{H}_n| < 0$

We have **negative definiteness (a maximum)** if $|\bar{H}_2| > 0, |\bar{H}_3| < 0, |\bar{H}_4| > 0, \dots$ etc.

Summary

Condition	Max	Min
F.O.C.	$Z_1 = \dots = Z_n = 0$	$Z_1 = \dots = Z_n = 0$
S.O.C.	$ \bar{H}_2 > 0, \bar{H}_3 < 0, \bar{H}_4 > 0, \dots, (-1)^n \bar{H}_n > 0$	$ \bar{H}_n < 0$ for all n

5 Utility Maximization and Consumer Demand

A classic problem in microeconomics supposes you have a utility function that depends on two goods x, y whose prices p_x and p_y are exogenously determined. Suppose a consumer has budget income I . The problem can be stated as follows

$$\max_{x,y} U = u(x, y) \quad \text{s.t.} \quad p_x x + p_y y = I$$

Set up the first order condition using the Lagrangian

$$L = u(x, y) + \lambda[I - p_x x - p_y y]$$

$$\mathcal{L}_\lambda : I - p_x x - p_y y = 0 \quad (1)$$

$$\mathcal{L}_x : U_x - \lambda p_x = 0 \quad (2)$$

$$\mathcal{L}_y : U_y - \lambda p_y = 0 \quad (3)$$

Using (2) and (3) we get $\frac{U_x}{p_x} = \frac{U_y}{p_y} = \lambda$.

This is a classic result from consumer theory: in order to maximize utility, consumers must equalize the ratio of marginal utility to price. The additional benefit per dollar for one good equals the additional benefit per dollar of the other good.

As before, $\lambda^* = \frac{\partial U^*}{\partial I} \Rightarrow$ the marginal utility of an additional dollar (i.e., relaxing the constraint).

5.1 Aside: Total Derivatives

Let's take an aside and consider total derivatives. Consider a function $y = f(x, w)$ where $x = g(w)$. Thus, $y = f(g(w), w)$. The three variables y, x , and w are all related. Here, w affects y through two separate channels:

- *Indirectly*, via the function g and then f , $f(g(w), w)$
- *Directly*, via the function f .

The direct effect can be represented with f_w or $\frac{\partial y}{\partial w}$. The indirect effect can be expressed by the product $f_x \frac{dx}{dw}$ (using the chain rule) since it is a composite function.

If we add up the two effects we get the **total derivative**

$$\frac{dy}{dw} = f_x \frac{dx}{dw} + f_w$$

We can also multiply both sides by dw

$$dy = f_x dx + f_w dw \Rightarrow \text{total differential}$$

Example: Find the total derivative of $y = f(x, w) = 3x - w^2$ where $x = g(w) = 2w^2 + w$.

We have $f_x = 3$, $\frac{dx}{dw} = (4w + 1)$, $f_w = -2w$

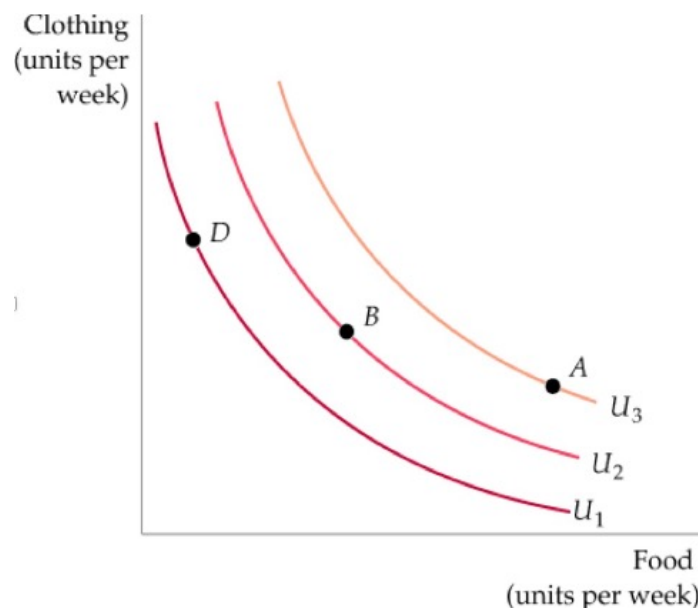
$$\Rightarrow \frac{dy}{dw} = f_x \frac{dx}{dw} + f_w = 3(4w + 1) + (-2w)$$

5.2 Indifference Curves and the MRS

Given that we can express the utility function mathematically we can plot a picture—an *indifference curve* (IC). ICs are defined as the locus of the combinations of x and y that will yield a constant level of U . On an IC, $du = U_x dx + U_y dy = 0$

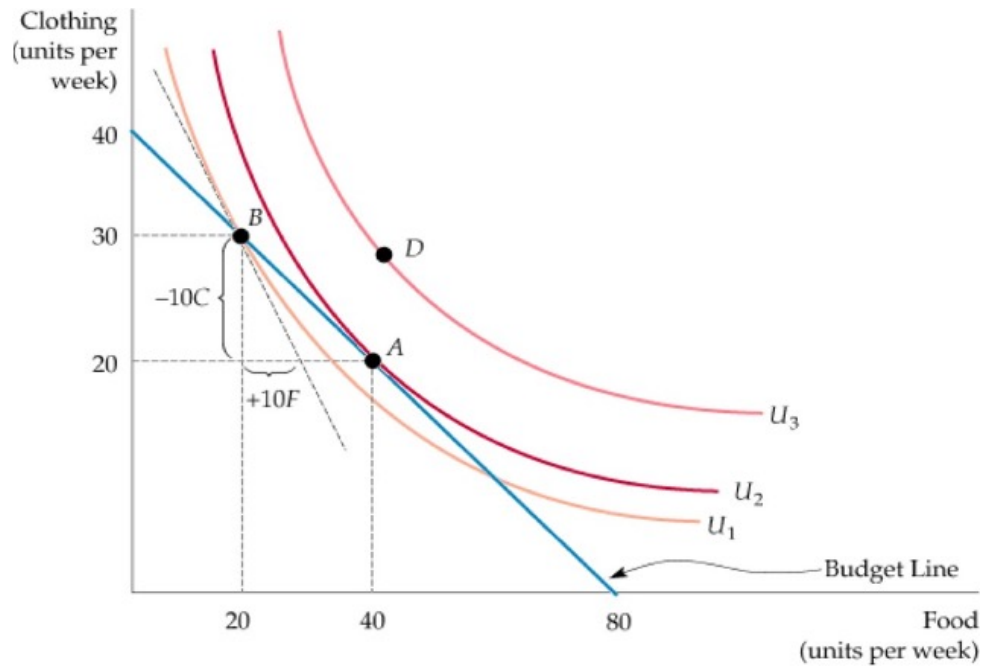
$$\Rightarrow \frac{dy}{dx} = \frac{-U_x}{U_y}$$

$\frac{dy}{dx} = \frac{-U_x}{U_y}$ is the slope of an indifference curve. We call the slope of an IC the marginal rate of substitution (MRS). The MRS measures the value that an individual places on an extra unit of a good in terms of another. Which curve is preferred by a consumer?



We can also plot the budget line $p_x x + p_y y = I$, which can be rewritten as

$$y = \frac{I}{p_y} - \frac{p_x}{p_y} x$$



Which is the most preferred point? A , B , or D ? What is the relationship between the slope of the budget line and the marginal rate of substitution at point B ?

To maximize utility, a consumer allocates their budget such that the slope of the budget line equals the slope of the ICs.

$$\frac{U_x}{U_y} = \frac{p_x}{p_y}$$

Point A in this case.

5.3 Second Order Condition

If the bordered hessian in this instance is positive $\implies$ a max

$$|\bar{H}| = \begin{vmatrix} 0 & p_x & p_y \\ p_x & U_{xx} & U_{xy} \\ p_y & U_{yx} & U_{yy} \end{vmatrix}$$

$$= 2p_x p_y U_{xy} - p_y^2 U_{xx} - p_x^2 U_{yy} > 0 \implies \text{negative definite}$$